Smart Token and Queue Management System for Campus Services

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1. Abstract

Waiting in long, chaotic lines at campus service points—like the canteen, library, or Xerox shop—is a daily struggle for students. It wastes precious break time, causes unnecessary stress, and makes things harder for both students and shopkeepers. Our project aims to change that. We're proposing a mobile-based Smart Token and Queue Management System that automates token generation, lets students book slots during their break hours, supports both cash and digital payments, and provides live updates on token status. With this system, we hope to make campus life smoother, faster, and a lot less stressful for everyone.

2. Introduction

Think about the last time you tried to grab a quick snack during your break, only to find yourself stuck in a long, unorganized queue. Or maybe you needed to pay your library dues or print some notes, but the crowd made it nearly impossible to get things done on time. These are everyday problems for students across our campus—at the Main Canteen, Snacks Bar, library counters, stationery shops, and more.

We believe there's a better way. Our solution is a unified, mobile-based system that brings together digital payments, slot booking during break hours, real-time token displays, and an easy-to-use interface for shopkeepers. The goal? Less waiting, less chaos, and a much better campus experience.

3. Objectives

- **Let students pre-book tokens** during their assigned break hours, so they don't have to wait in line.
- **Support both digital and cash payments** within a single app.
- **Give shopkeepers a simple dashboard** to manage orders and update token status.
- **Display live token numbers** on screens and online, so everyone knows when it's their turn.
- **Organize queues smartly** based on order complexity and priority.

4. What's Wrong with the Current System?

- Long, messy queues everywhere.
- No way to know your place in line or how long you'll wait.

- Payments are often cash-only and unorganized.
- Crowds get out of control during breaks.
- Both students and shopkeepers end up stressed and frustrated.

5. Our Solution: How the System Works

We're building a mobile app for both students and shopkeepers. Here's how it works:

- **Students** log in with their register number and can book tokens only during their allotted break slots. They pay through the app or in cash at the counter, get a token ID, and can check their queue status anytime.
- **Shopkeepers** use a dashboard to see bookings, mark orders as ready, and notify students.
- **Live token status** is displayed on campus TVs and a web panel, so everyone knows what's happening.
- **Token assignment is smart:** pre-made items are served faster, while made-to-order items follow the queue.

6. Key Features

- **Student App:** Book tokens, make payments, view tokens, get receipts.
- **Shopkeeper Panel:** Accept or decline orders, manage the queue, mark orders as ready.
- **Admin Interface:** Add or edit shops, manage users.
- **Live Token Display:** For everyone to see—on TV screens or online.
- **Notifications:** Get alerts when your order is ready.
- **Payment Integration:** Supports Razorpay, UPI, and more.

7. Break Hour Rules

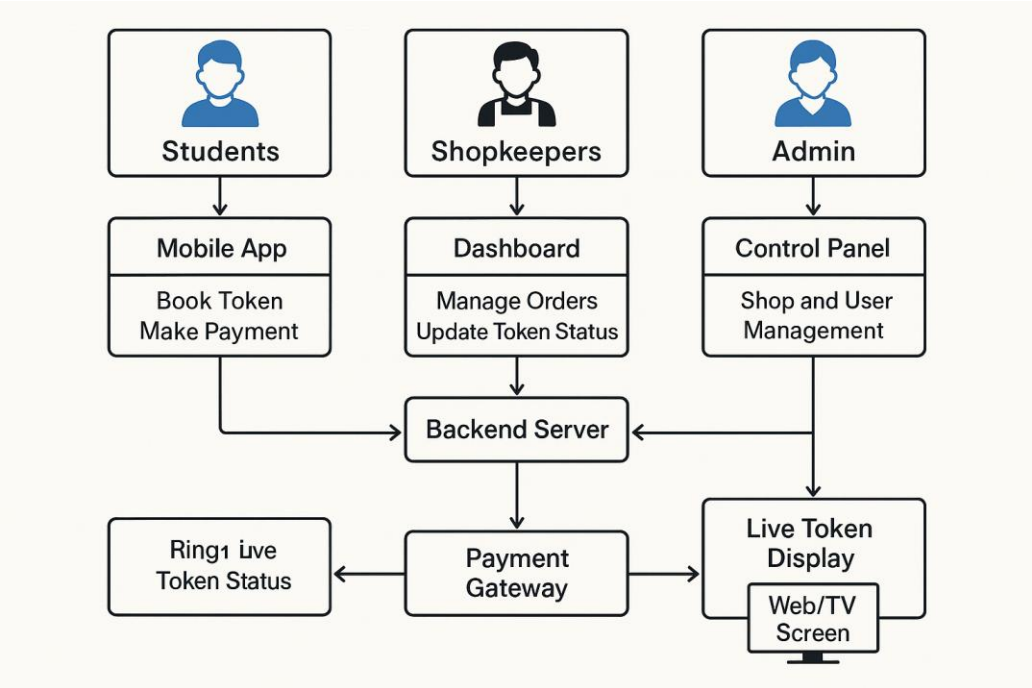
To keep things fair, only students in their designated break slot can book tokens.
Example:

- 1st Year: 08:50–09:20
- 2nd Year: 09:30–10:00
- 3rd Year: 10:30–11:00
  (Shift 2 follows a similar structure.)

8. Technology Stack

- **Frontend:** Flutter or Android (Java/Kotlin)
- **Backend:** Firebase or Node.js with MySQL
- **Database:** Firebase Realtime DB or Firestore
- **Authentication:** Register number-based
- **Payment:** Razorpay or PhonePe SDK
- **Display:** Web interface + HDMI-enabled TVs

9. System Architecture



10. Implementation Timeline

Week	Task
1	System Design & Requirements Finalization
2	Frontend UI Design (Student + Shopkeeper)
3	Backend Setup + Authentication
4	Token Logic + Break Slot Rules
5	Payment Gateway Integration
6	Token Display Module (Web/TV)
7	Testing + Feedback from Stakeholders

Week	Task
8	Final Deployment + Documentation

11. Benefits & Outcomes

- Shorter wait times for students.
- Better crowd control and less chaos.
- Easier payments and receipts.
- Less manual work for shopkeepers.
- Pushes our campus toward a digital future.

12. Future Scope

- RFID-based automatic token pickup.
- Integration with the existing student portal.
- Expandable for use at campus events and labs.

13. Conclusion

Our Smart Token and Queue Management System is a simple, practical fix for a problem we all face every day. By making queues digital and payments hassle-free, we're making campus life better for everyone—students and shopkeepers alike. It's a step toward a more organized, efficient, and digital campus.